Orthogonality and distances

Lecture and worked examples

Maths Explorer

Objectives

Calculate a dot product; justify an orthogonality; find an orthogonal projection and a distance; calculate a volume.

Course notes

The coordinate calculations in this chapter are carried out in an orthonormal frame of space.

Dot product in space

For non-zero vectors $\vec{u}, \vec{v}$, $\vec{u} \cdot \vec{v} = \|\vec{u}\|\|\vec{v}\| \cos\theta$, where $\theta \in [0, \pi]$ is their angle. If one of them is zero, the dot product is 0.

$$\vec{u} \cdot \vec{v} = \frac{1}{2}(\|\vec{u} + \vec{v}\|^2 - \|\vec{u}\|^2 - \|\vec{v}\|^2).$$

In an orthonormal frame, for $\vec{u} = (x, y, z)$ and $\vec{v} = (x', y', z')$,

$$\vec{u} \cdot \vec{v} = xx' + yy' + zz'.$$

The dot product is symmetric and bilinear; $\vec{u} \cdot \vec{u} = \|\vec{u}\|^2$. Two vectors are orthogonal if and only if their dot product is zero.

Dot product in space

$\vec{u} = (1, 2, 2)$ and $\vec{v} = (2, -1, 0)$ are orthogonal: $2 - 2 + 0 = 0$. Their norms are 3 and $\sqrt{5}$.

Line, plane and normal vector

Two lines are orthogonal if their direction vectors are. A line is orthogonal to a plane if it is orthogonal to two non-collinear directions of that plane. A normal vector to a plane is a non-zero vector orthogonal to all its directions.

Line, plane and normal vector

If $\vec{n} \cdot \vec{u} = \vec{n} \cdot \vec{v} = 0$, then, for every $\vec{w} = s\vec{u} + t\vec{v}$, $\vec{n} \cdot \vec{w} = s\vec{n} \cdot \vec{u} + t\vec{n} \cdot \vec{v} = 0$.

Line, plane and normal vector

If A belongs to the plane $\mathcal{P}$ with normal vector $\vec{n}$, then $M \in \mathcal{P}$ if and only if $\overrightarrow{AM} \cdot \vec{n} = 0$.

Orthogonal projection and distance

The orthogonal projection H of M onto a plane is the unique point of the plane such that $\overrightarrow{MH}$ is normal to the plane; if M belongs to the plane, $H = M$. The distance from M to the plane is MH. For a line with direction vector $\vec{u}$, the projection satisfies $H \in d$ and $\overrightarrow{MH} \cdot \vec{u} = 0$.

Orthogonal projection and distance

For every point P of the plane, $\overrightarrow{MH} \perp \overrightarrow{HP}$, so $MP^2 = MH^2 + HP^2 \geq MH^2$. Equality holds only for $P = H$. The projection is the closest point of the plane.

Orthogonal projection and distance

If $d = A + \mathbb{R}\vec{u}$, write $H = A + t\vec{u}$ and solve $(A - M + t\vec{u}) \cdot \vec{u} = 0$. Hence $t = \dfrac{\overrightarrow{AM} \cdot \vec{u}}{\|\vec{u}\|^2}$.

Angles, areas and volumes

For an angle $\widehat{ABC}$, use $\overrightarrow{BA}$ and $\overrightarrow{BC}$: $\cos\widehat{ABC} = \dfrac{\overrightarrow{BA}\cdot\overrightarrow{BC}}{BA \times BC}$. For the acute angle between two lines, take the absolute value of the quotient. The area of a triangle is $\dfrac{bh}{2}$; the volume of a tetrahedron is $\dfrac{\mathcal{A}_{\text{base}}h}{3}$, where h is an orthogonal distance. The units are squared and cubed respectively.

Expansions and loci

By bilinearity,

$$\|\vec{u}+\vec{v}\|^2 = \|\vec{u}\|^2 + 2\vec{u}\cdot\vec{v} + \|\vec{v}\|^2, \qquad \|\vec{u}-\vec{v}\|^2 = \|\vec{u}\|^2 - 2\vec{u}\cdot\vec{v} + \|\vec{v}\|^2.$$

So we can also calculate $\vec{u}\cdot\vec{v} = \frac{1}{2}(\|\vec{u}\|^2 + \|\vec{v}\|^2 - \|\vec{u}-\vec{v}\|^2)$. For two distinct points A, B with midpoint I, the locus of points M such that $MA = MB$ is the plane through I with normal vector $\overrightarrow{AB}$: the *perpendicular bisector plane*. Indeed, writing the vectors from I, the expansions give $MA^2 - MB^2 = 2\overrightarrow{IM}\cdot\overrightarrow{AB}$. Equality of the distances is therefore equivalent to $\overrightarrow{IM}\cdot\overrightarrow{AB} = 0$. **Extension, not specifically assessed in 2026-2027.** Two planes are perpendicular when their normal vectors are orthogonal. This property must not be confused with parallelism of planes, which corresponds to collinear normal vectors.

Research activity

The closest point on a line - 35 min.

In an orthonormal frame of space, consider $A(1;0;0)$, $M(0;2;3)$ and the line d through A with direction vector $\vec{u} = (1;1;0)$. We look for the point of d closest to M.

Research activity (2)

1. Express, in terms of a real parameter t, the coordinates of a point H of d.
2. Find the value of t for which $\overrightarrow{MH} \cdot \vec{u} = 0$. Deduce the coordinates of the resulting point H.
3. Calculate the distance MH.
4. Let P be any point of d. Prove that $MP^2 = MH^2 + HP^2$, then deduce that H is the only point of d closest to M.